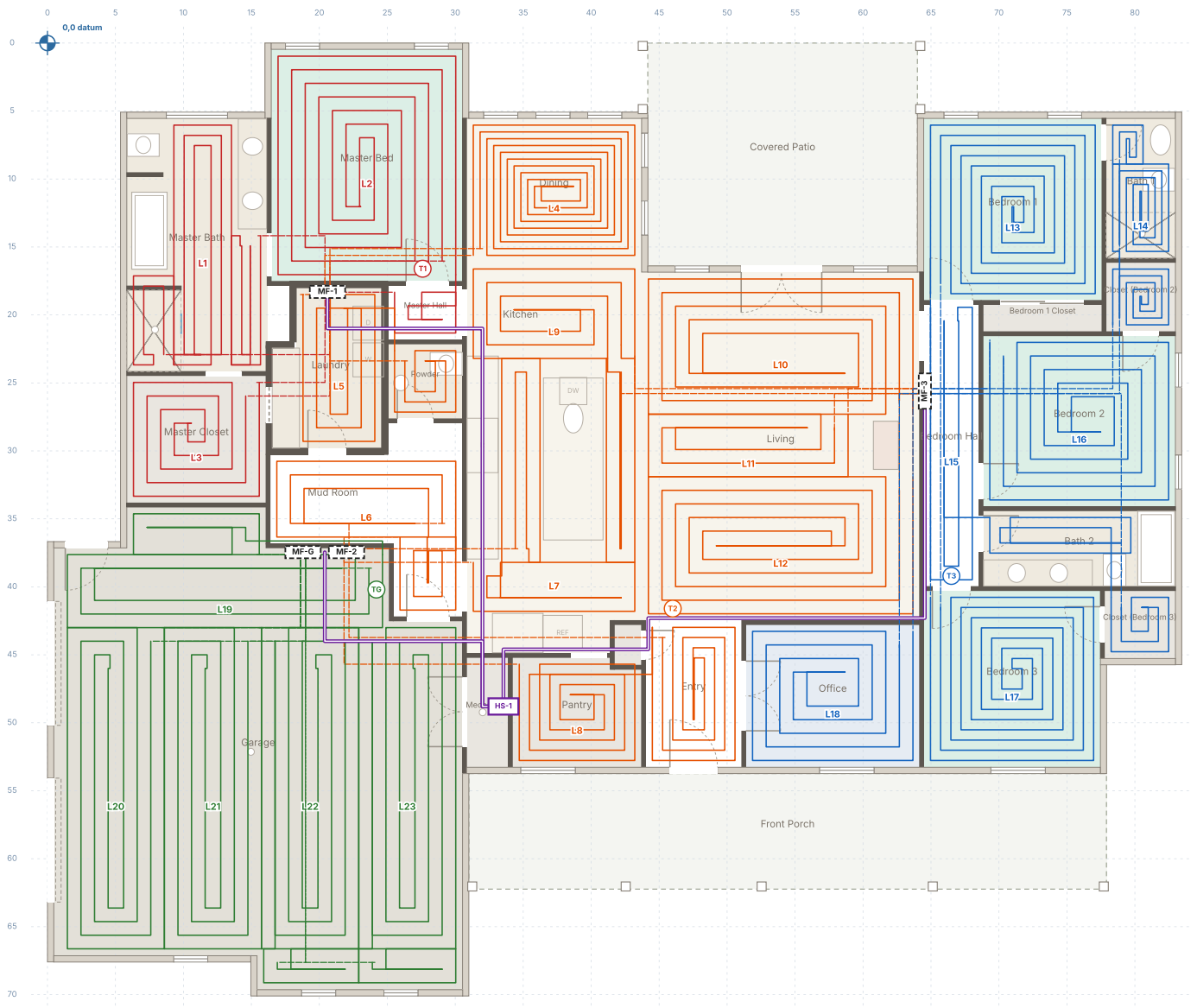
REAR · PATIO




FRONT · PORCH

Zone 1, Master suite (T1) · Zone 2, Great room and utility (T2) · Zone 3, Bedroom wing and office (T3) · Zone G, Garage (held at 50°F) (TG)

leader pair (drawn spread out) · 1" feed pair from HS-1 · manifold

23 loops, 4,922' of 1/2" PEX, 4 manifolds. · Red zone 1, orange zone 2, blue zone 3, green garage. · 54 kBtu/h at design, 110°F supply. · Zones are thermostats on loop actuators, not manifolds.

draft

# R-1 Radiant · 64 Stone Sheep Circle

PEX loops in the slab, four thermostat zones, manifolds and flows at Powell, WY design (-11°F). App version v14. From the owner's plans (Larson House, 6/11/26).

## Notes

1. Positions are taped from the nearest wall face, exterior walls first. Every panel stops 6" off the walls, so interior wall plates fasten in that band; mark every tube crossing under a wall line on the vapor barrier and keep shots and anchors 12" from it.
2. Tube: 1/2" O2-barrier PEX in counterflow spirals (supply in, return out between), tied to the mesh at mid-depth of a 4" slab. 8 loops run at 9" o.c. or 6" (baths, and rooms needing 15 Btu/h per sq ft or more; 6" in the dining room and Bath 1), the rest at 12". One continuous coil per loop, no fittings in the slab. Tag both ends of every loop at its manifold.
3. No tube under the kitchen base cabinets, island, fridge or back cabinet, the tubs, the vanities, the laundry cabinets or the fireplace hearth. Keep tube 6" from toilet flanges and 12" from floor and shower drains; the showers' floors are heated.
4. Under the slab: R-10 (2") rigid foam under the whole heated slab and the garage slab, with R-10 at the slab edge (IECC 2021 zone 6 for a heated slab), and a 10-mil vapor barrier over the foam.
5. Zoning is by thermostat: four zones (master suite, great room and utility, bedroom wing and office, garage) on 24 V actuators at each loop's manifold port, so a room can get its own thermostat later by rewiring, without new tube.
6. Manifolds, three stations: MF-1 (laundry, on the north wall between the linen alcove and the dryer) serves Master Bed, Master Bath, Master Closet, Master Hall, Laundry, Powder, Dining; MF-2 (garage, on the mud room wall, beside MF-G) serves Mud Room, Kitchen, Pantry, Entry; MF-G (garage, on the mud room wall, beside MF-2) serves Garage; MF-3 (bedroom hall, recessed in the west wall north of the fireplace) serves Kitchen, Living, Office, Bedroom Hall, Bedroom 1, Bath 1, Closet (Bedroom 2), Bedroom 2, Bath 2, Bedroom 3, Closet (Bedroom 3). HS-1 in the mechanical closet feeds all three.
7. Leaders run in the slab as tight pairs (drawn spread out). Sleeve them in 3/8" foam where they pass under another zone (136' in all) and wherever they cross a sawcut control joint. Feed pairs from HS-1 to

## Open items

- L14 (Closet (Bedroom 2), Bath 1): the floor carries all but about 135 Btu/h of the room at -11°F with 110°F water. Add a small electric towel warmer, or run the boiler hotter on the coldest days.
- Boiler model: HS-1 is a condensing boiler on natural gas or propane (decided 2026-10-13). Pick a combi or a boiler with an indirect tank if it should also make the domestic hot water and replace P-1's 50 gal heater; propane needs the boiler's LP conversion kit.
- Garage: kept at 50°F on its own manifold, pump and thermostat. Decide whether it runs a glycol mix behind a plate heat exchanger, which protects it if the house loses power in a cold snap with the doors open.
- Flooring: the loops assume hard-surface floors. Carpet in a bedroom needs about 10°F warmer water for that loop, or 9" spacing there.
- Mechanical closet: HS-1, pump P-R1 and the zone controller share it with P-1's water manifold and water heater; that is why no manifold is drawn there (O5 in the options shows the fourth manifold saves little tube). A combi boiler removes the water heater and makes room.
- MF-3 cabinet depth in the hall's 2×4 west wall (S-1), and the boiler, pump and zone controller circuits (E-1).

the three stations are 1" PEX, insulated end to end, 122' each way in all.

8. Flow: 10.9 gpm with every loop calling, each loop set on its flow meter at a 10°F drop. Design heat 54,493 Btu/h, of which 10,834 is the garage at 50°F; 10% is added for heat lost down and out the edge.
9. Water: 110°F supply at -11°F outside, reset lower as it warms, so any low-temperature source can run it. Floor surfaces stay at or under 90°F. At that supply L14 (Closet (Bedroom 2), Bath 1) falls about 135 Btu/h short at design; see the open items.
10. Pump: the hardest loop is L4 at 8.0 ft in the tube; add about 3 ft for the feed pair, manifold and actuator. A constant-ΔP ECM circulator at about 12 ft covers it.
11. Pressure test every loop at 100 psi water or 60 psi air for 24 hours before the pour and keep it under pressure through the pour. Photograph the layout with a tape in frame before concrete.

## Loop schedule

Leader is each way. Heat includes the share lost downward. Pick a row to find the loop.

| Loop | Zone | Rooms | Manifold | Spacing | In slab | Leader | Total | Btu/h | Flow | Head | Floor |
|---|---|---|---|---|---|---|---|---|---|---|---|
| L1 | 1 | Master Bath | MF-1 | 9" | 196' | 15' | 226' | 3,800 | 0.76 gpm | 5.5 ft | 85°F |
| L2 | 1 | Master Hall, Master Bed | MF-1 | 12" | 264' | 10' | 284' | 3,182 | 0.64 gpm | 5.1 ft | 76°F |
| L3 | 1 | Master Closet | MF-1 | 12" | 94' | 15' | 124' | 1,499 | 0.30 gpm | 0.6 ft | 79°F |
| L4 | 2 | Dining | MF-1 | 6" | 248' | 16' | 280' | 4,149 | 0.83 gpm | 8.0 ft | 87°F |
| L5 | 2 | Laundry, Powder | MF-1 | 12", 9" | 114' | 8' | 131' | 1,159 | 0.23 gpm | 0.4 ft | 77°F |
| L6 | 2 | Mud Room | MF-2 | 12" | 123' | 9' | 141' | 1,360 | 0.27 gpm | 0.6 ft | 76°F |
| L7 | 2 | Kitchen | MF-2 | 12" | 116' | 13' | 142' | 1,097 | 0.22 gpm | 0.4 ft | 76°F |
| L8 | 2 | Pantry, Entry | MF-2 | 9" | 205' | 27' | 260' | 2,930 | 0.59 gpm | 4.0 ft | 81°F |
| L9 | 2 | Kitchen | MF-3 | 12" | 160' | 26' | 211' | 1,412 | 0.28 gpm | 0.9 ft | 76°F |
| L10 | 2 | Living | MF-3 | 12" | 212' | 5' | 221' | 2,760 | 0.55 gpm | 3.1 ft | 76°F |
| L11 | 2 | Living | MF-3 | 12" | 90' | 10' | 110' | 963 | 0.20 gpm | 0.3 ft | 76°F |
| L12 | 2 | Living | MF-3 | 12" | 232' | 10' | 253' | 2,796 | 0.56 gpm | 3.6 ft | 76°F |
| L13 | 3 | Bedroom 1 | MF-3 | 9" | 225' | 10' | 246' | 3,766 | 0.75 gpm | 5.9 ft | 81°F |
| L14 | 3 | Closet (Bedroom 2), Bath 1 | MF-3 | 6" | 137' | 27' | 191' | 2,161 | 0.43 gpm | 1.7 ft | 90°F |
| L15 | 3 | Bedroom Hall, Bath 2, Closet (Bedroom 3) | MF-3 | 12", 9" | 188' | 20' | 229' | 2,371 | 0.47 gpm | 2.5 ft | 80°F |

| Loop | Zone | Rooms | Manifold | Spacing | In slab | Leader | Total | Btu/h | Flow | Head | Floor |
|---|---|---|---|---|---|---|---|---|---|---|---|
| L16 | 3 | Bedroom 2 | MF-3 | 12" | 175' | 10' | 196' | 2,599 | 0.52 gpm | 2.5 ft | 78°F |
| L17 | 3 | Bedroom 3 | MF-3 | 9" | 214' | 18' | 251' | 3,480 | 0.70 gpm | 5.3 ft | 81°F |
| L18 | 3 | Office | MF-3 | 12" | 131' | 22' | 174' | 2,175 | 0.43 gpm | 1.6 ft | 79°F |
| L19 | G | Garage | MF-G | 12" | 195' | 7' | 209' | 1,912 | 0.38 gpm | 1.5 ft | 56°F |
| L20 | G | Garage | MF-G | 12" | 196' | 19' | 234' | 2,118 | 0.42 gpm | 2.1 ft | 56°F |
| L21 | G | Garage | MF-G | 12" | 196' | 12' | 219' | 2,121 | 0.42 gpm | 1.9 ft | 56°F |
| L22 | G | Garage | MF-G | 12" | 248' | 22' | 292' | 2,341 | 0.47 gpm | 3.1 ft | 56°F |
| L23 | G | Garage | MF-G | 12" | 248' | 24' | 297' | 2,342 | 0.47 gpm | 3.1 ft | 56°F |
| Zone 1 | 1 | Master suite | MF-1 | | 3 loops | | 633' | 8,482 | 1.7 gpm | | |
| Zone 2 | 2 | Great room and utility | MF-1, MF-2, MF-3 | | 9 loops | | 1,750' | 18,626 | 3.7 gpm | | |
| Zone 3 | 3 | Bedroom wing and office | MF-3 | | 6 loops | | 1,287' | 16,551 | 3.3 gpm | | |
| Zone G | G | Garage (held at 50°F) | MF-G | | 5 loops | | 1,252' | 10,834 | 2.2 gpm | | |

## Manifolds

| Tag | Loops | Where | Zones | Feed |
|---|---|---|---|---|
| MF-1 | 5 loops: L1, L2, L3, L4, L5 | Laundry, on the north wall between the linen alcove and the dryer | 1, 2 | 1" pair F-1, 43' each way |
| MF-2 | 3 loops: L6, L7, L8 | Garage, on the mud room wall, beside MF-G | 2 | 1" pair F-2, 24' each way |
| MF-G | 5 loops: L19, L20, L21, L22, L23 | Garage, on the mud room wall, beside MF-2 | G | 1" pair F-2, 24' each way |
| MF-3 | 10 loops: L9, L10, L11, L12, L13, L14, L15, L16, L17, L18 | Bedroom hall, recessed in the west wall north of the fireplace | 2, 3 | 1" pair F-3, 54' each way |

## Manifold options

The same loops routed to each layout, every loop to its nearest station (two 12-port manifolds at most per station). Leaders count both ends and the 2' rise at the manifold. ★ is drawn.

| Option | Layout | 1/2" PEX | Leaders | Longest / shortest loop | Loops over 300' | Leader under another zone | Feed pairs | Stations |
|---|---|---|---|---|---|---|---|---|
| O1 | All manifolds in the mechanical closet, beside the heat source | 5,763' | 1,572' | 354' / 143' | 3 | 420' | none | 1 |
| O2 | Owner's first sketch: laundry corner, two on the garage side of the mud room wall, two in the bedroom hall beside the fireplace | 4,918' | 718' | 302' / 108' | 1 | 105' | 122' × 2 | 3 |
| O3 | Two stations: the mechanical closet, and the bedroom hall | 5,226' | 1,029' | 334' / 110' | 2 | 256' | 54' × 2 | 2 |
| O4 ★ | Three stations: laundry north wall (MF-1), mud room wall in the garage (MF-2 and MF-G), bedroom hall north of the fireplace (MF-3) | 4,928' | 719' | 294' / 110' | 0 | 107' | 122' × 2 | 3 |
| O5 | O4 plus a fourth manifold in the mechanical closet for the kitchen, pantry and entry | 4,883' | 672' | 292' / 110' | 0 | 89' | 122' × 2 | 4 |

## Bill of materials

Rough-order quantities from the drawing geometry, for pricing. The installer takes off the final list.

| Item | Qty | Notes |
|---|---|---|
| 1/2" O2-barrier PEX, 1,000' coils | 6 coils | 4,922' + 5% waste; one length per loop, no splices |
| 1" PEX supply + return, insulated | 244 lf | HS-1 to the three manifold stations, under the slab |
| Stainless manifolds with flow meters | 4 | MF-1 5-loop, MF-2 3-loop, MF-G 5-loop, MF-3 10-loop |
| 24 V thermal actuators | 23 | one per loop |
| Zone controller + thermostats | 1 + 4 | T1, T2, T3, TG; outdoor reset sensor |
| Circulators | 2 | P-R1 constant ΔP for the house, P-RG for the garage with its mixing valve |
| Manifold cabinets | 4 | recessed at MF-3, surface at MF-1 (sound wall), MF-2 and MF-G |
| R-10 rigid foam, 2" | 4,167 sf | under the heated slab and the garage, plus the slab edge |
| 10-mil vapor barrier | 4,584 sf | over the foam, lapped and taped |
| Foam sleeve 3/8" for 1/2" PEX | 189 lf | leaders under other zones and at control joints |
| PEX staples / ties, 18" o.c. | 3.3k | to the mesh or foam |